The Effect of Forest Age on Machine Learning Evapotranspiration Generalisation

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ABSTRACT

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Key words: Evapotranspiration, Forest Age, Machine Learning, FLUXNET.
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A dissertation submitted to the University of Bristol in accordance with the requirements of the degree of Master of Science by advanced study in Bioinformatics in the Faculty of Health and Life Sciences.

Declaration

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgement. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn
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Contents

Introduction

Evapotranspiration

Terrestrial evapotranspiration (ET) is a key component of the global hydrological cycle describing the transition of water from liquid to gas at Earth's surface and further transfer to the atmosphere, representing a crucial element of the energy and water exchange between terrestrial ecosystems and the climate system. The term emphasises the combination of water vapourisation flux from (1) abiotic evaporation from soil and plant canopy surfaces, and (2) biotic transpiration through stomata to the atmosphere (Katul *et al.* 2012). Approximately 60% of precipitation on the land surface is returned to the atmosphere through ET (Oki and Kanae 2006), and the associated latent heat flux, λET (where λ is the latent heat of vaporisation), accounts for 50% of the solar radiation absorbed by the land surface (Trenberth *et al.* 2009).

The contribution of the variables which make up ET is not equal, with Lian *et al.* (2018) estimating a global T/ET ratio of 0.62$\pm$0.06, indicating plant transpiration as the leading component of ET worldwide. The difference between precipitation and ET ($P-ET$) represents the source of available freshwater content which feeds rivers, lakes, and groundwater discharge (Swenson and Wahr 2006). Given the importance of accounting for freshwater availability and the rigorous measurements of P already available, estimating ET becomes an important metric for predicting regional food supplies, energy generation, human and ecosystem health, social unrest, and in assessing the overarching patterns of global climatic shifts (Rodell *et al.* 2018; Amani and Shafizadeh-Moghadam 2023; Xue *et al.* 2025). In turn, these estimates can steer regulations and policies in climate science and water resource management, representing a critical link between the quantification of terrestrial water fluxes and land systems governance (Haiyang Shi *et al.* 2024).

Estimating Evapotranspiration

The direct measurement of ET is the most accurate and uses lysimeters, eddy covariance (EC), or sap flow, but these are costly to set up and only measure at the plot scale, resulting in spatially constrained measurements which makes them unsuitable for describing ET at regional scales and beyond (Amani and Shafizadeh-Moghadam 2023; Liyew *et al.* 2025). The uneven distribution and low sampling rate of EC data in ecoregions of Africa, Oceania (excluding Australia), and South America may be attributed to the costs associated with these methods, despite efforts to mitigate this (Hill *et al.* 2017; Markwitz and Siebicke 2019). As a result of these issues, methods to estimate ET indirectly in areas without the capacity for direct measurement have been derived, including: (1) traditional process-based models which rely on pre-defined equations such as the Penman-Monteith (PM) and Hargreaves-Samani (HS) equations (Monteith 1965; Hargreaves and Samani 1985); and more recently (2) machine learning (ML) methods such as Artificial Neural Networks (ANN) and Random Forest (RF) models trained on historical data (T. Xu *et al.* 2018). Remote sensing (RS) -based models represent a distinct category as these are typically formed from process-based and ML models which integrate RS data, for example the SEBS model (Su 2002). The development of these methods has permitted global ET estimates to be produced using more easily acquired and widely measured variables such as air temperature and relative humidity (S. Pan *et al.* 2020), yet significant uncertainties remain, particularly in underrepresented regions and ecosystems with sparse EC coverage, as well as in complex or heterogeneous landscapes (T. Xu *et al.* 2018; Amani and Shafizadeh-Moghadam 2023).

Forest Age and Evapotranspiration dynamics

Canopies and Conductance in Ageing Forests

The magnitude of ET is governed by many environmental factors such as solar radiation, leaf area index (LAI), air temperature, vapour pressure deficit (VPD), soil moisture, and wind speed (Yang *et al.* 2023). Of these, Lian *et al.* (2018) identified LAI as the primary driver of T/ET spatial patterns, with Kool *et al.* (2014) suggesting that in systems with full canopy cover ET is often assumed to be "similar enough to T to permit correlation of ET with biomass". However, this assumption may break down in densely vegetated systems with multi-layer canopies such as forests. In these systems, saturation issues with the remote measurement of LAI lead to issues in capturing the vertical canopy complexity in full, leading to a loss of information which drives rainfall interception and subsequent ET dynamics (Q. Wang *et al.* 2022).

Hardiman *et al.* (2013) demonstrate that while LAI may be stable in stands older than 50 years, canopy complexity continues to increase with age through >160 years of stand development and had a greater impact on annual net primary productivity (NPP) and overall resource use efficiency than LAI across all stands, suggesting vegetation indices may saturate before canopy complexity peaks in forests. Oishi *et al.* (2020), in a chronosequence study on broadleaf forests in the US, report that interception of incoming rainfall strongly increases with age from ∼2% (12 year old stand) to 20% (200 year old stand) despite only small differences in LAI, an effect in part due to branch architecture which varies with age. This study also reported that annual canopy transpiration (T) was over twice as high in the 35 year old stand compared to the 12 and 200 year old stands, with increasing canopy interception (and subsequent evaporation) in the oldest stand largely offsetting the reduction in T such that total ET was similar between the two older stands, but remained lower in the youngest stand (Figure 1). While LAI remains an important structural parameter for informing ET predictions and quantifying biological processes relating to vegetation dynamics, these findings together suggest that age-driven changes in canopy complexity and architecture introduce ET dynamics that may persist beyond the saturation point of RS vegetation indices.

Alongside canopy architecture, stomatal conductance (g_s) declines as trees grow taller due to gravitational reductions in leaf water potential and decreasing hydraulic conductance with increasing path length from soil to leaf (Ryan and Yoder 1997). For example, Schäfer *et al.* (2000) demonstrated that the mean stomatal conductance of tree crowns decreased by approximately 60% over a 30m increase in tree height, and Hubbard *et al.* (1999) established a 44% drop in hydraulic conductance in old trees compared to young trees in a mixed age stand. As g_s directly influences T (Monteith 1965), these age related changes introduce further complexity into ET dynamics and suggest that age could be a key source of ET variability across forest ecosystems.

Disturbances and Climate Change

Forest age distributions are increasingly being affected by climate change-driven disturbances - such as fire, drought, pest attack, and pathogens - which are expected to continue in the coming decades (Solomon *et al.* 2009; Seidl *et al.* 2017; Altman *et al.* 2024). Some of these disturbances are able to completely change forest demography by causing widespread mortality (Dale *et al.* 2001), effectively resetting the age of the forest landscape. Land use changes also directly affect forest age distributions, for example the recovery of deforested areas in southern China following the implementation of forestation policies has resulted in a complex pattern of forest age structure in recent years, increasing average tree cover from 21% in 1987 to 38% in 2016 and leading to large areas now being dominated by young forests (Tong *et al.* 2020; Leng *et al.* 2024), with similar patterns of young forests due to recent disturbance found worldwide (Besnard *et al.* 2021).

As forests are exhibiting greater age heterogeneity, and with climate change increasing the severity and frequency of disturbances worldwide, forest age may represent an increasingly important variable for ET estimation, capturing age-specific effects such as canopy complexity and hydraulic conductance limitations which are underrepresented in current ET estimations.

Machine Learning and Integrating Forest Age

Evapotranspiration Estimation Methods

There are many different methods for the estimation of ET as no single model performs reliably across all conditions and regions (Derardja *et al.* 2024), thus the data requirement can be significantly varied across models (Zhao *et al.* 2013). Process-based models define the relationship between input variables and ET explicitly for estimations which are developed through hydrological theory, yet often suffer from a requirement of comprehensive meteorological data as input, and their rigid structure can limit adaptability and generalisation (Amani and Shafizadeh-Moghadam 2023). Meanwhile, ML approaches do not explicitly define relationships, instead relying on the underlying architecture of models to optimise against the provided training data to derive the relationship between inputs and ET.

ML methods have been used extensively for ET estimation in recent years and have demonstrated high spatiotemporal prediction accuracy and a capacity to outperform process-based (S. Pan *et al.* 2020; Yan *et al.* 2023). The increased precision of ML ET estimations in data rich regions may be due to their ability to leverage both linear and non-linear relationships for mapping between input data and ET estimates due to their inherent lack of prior assumptions between the two (Simon Besnard *et al.* 2018; Xue *et al.* 2025). In data limited contexts, hybrid approaches combining ML and process-based models have shown improved performance over standalone methods, whether by using process-based outputs as ML input features or by using ML to estimate parameter values for process-based equations (Y. Liu *et al.* 2021; Yan *et al.* 2023). Traditional (non deep learning) ML approaches, such as RF and Support Vector Machine (SVM), are commonly used and also show good ET predictive performance, with T. Xu *et al.* (2018) comparing the capabilities of five ML models against EC data as ground truth, finding the RF model to slightly outperform the others in uncertainty, with all models predicting ET to high accuracy (R^2 0.87 – 0.89). Recurrent neural networks (RNNs) are a specialised form of ANN which are highly capable in extracting long term dependencies within time series data and handling noisy datasets, indicating strong potential for application using long-term EC datasets as training for ET estimations (Yan *et al.* 2023).

While the use of ML methods is promising in reducing uncertainties and improving ET estimation accuracy, they can struggle with extrapolation beyond training data and lack the interpretability of process-based models, often being termed ‘black boxes’ due to their inherent complexity and lack of decision making explainability (Hassija *et al.* 2024). The quality of ML ET estimations is highly dependent on the provided training data, model hyperparameter tuning, and choice of predictors (feature selection) - all of which relies on user knowledge and pre-existing data for the specific ecosystem in question (Amani and Shafizadeh-Moghadam 2023). Therefore careful consideration must be taken during ML training and evaluation to form a relevant and accurate ET estimation model.

Feature Selection and FLUXNET

The selection of features in ML ET estimation models can vary from a minimal data input (Ahmadi *et al.* 2024) to a broad choice of variables from multiple sources (Y. Liu *et al.* 2021). Table 2 presents examples on the range of input features included in ML models used for ET predictions. The range of selected input features is often constrained by the availability of meteorological data in the target region, with T. Xu *et al.* (2018) demonstrating that successively adding more input variables increased ET prediction accuracy and reduced uncertainty in five different ML models, suggesting that the inclusion of additional relevant features, such as forest age, could improve model performance. However, the number of input variables used may be lower in attempts to form models which can be applied to data limited regions for greater generalisation power (Ahmadi *et al.* 2024). In either case, access to standardised, high quality data is key in forming ML models for ET.

An important and commonly used resource to form ML ET models is the FLUXNET (Baldocchi *et al.* 2001; Pastorello *et al.* 2020), which is formed from data collected at flux towers worldwide and comprises gas and energy flux measurements within a variable footprint around the tower - including latent heat (λET) from which ET can be directly calculated, alongside meteorological and biological measurements at these sites. The standardisation and open access of FLUXNET data across sites enables confident comparisons and EC data access across multiple ecosystems, and yet despite the abundance of variables contained within the FLUXNET, forest-specific features such as stand age are rarely included at sites (Pastorello *et al.* 2020). Given the potential impact of forest age on ET described above and the widespread use of FLUXNET data in training ML ET models (Jung *et al.* 2010; Simon Besnard *et al.* 2018; R. Wang *et al.* 2022), this omission represents a gap in the feature space available to these models which potentially limits their ability to generalise across forests of varying successional stages and structural complexities.

Machine Learning for Age-Evapotranspiration Relationships

The flexibility of ML models to capture both linear and nonlinear relationships between predictor and target variables has previously been used to include forest age data, with Simon

Besnard *et al.* (2018) using a RF model with and without forest age as a predictor variable for net ecosystem productivity (NEP) estimations, finding strong support for a nonlinear relationship between NEP and forest age which demonstrates both the strength of this ML method in detecting these relationships and a capacity for including forest age as an input feature. This adaptability of ML methods to detect underlying patterns within complex datasets provides an avenue for the age-driven effects of forest age on ET estimations using FLUXNET data to be further explored.

Incorporating Forest Age

Vegetation age is not a standardised variable within the FLUXNET2015 dataset, with site level metadata varying considerably in the availability and precision of age-related information (Pastorello *et al.* 2020). This presents a difficult issue for investigating age-driven effects, particularly within ML models which require consistent and quantified inputs to derive relationships.

Simon Besnard *et al.* (2018) considered different scenarios recorded in databases and publications to define forest age, from complete stand replacement (age reset) to minor disturbances (age unchanged). These scenarios permitted age to be quantified at 126 FLUXNET sites, but the uncertainties involved with these calculations are large due to the qualitative and non-standardised reporting of disturbances, with particularly large uncertainties at sites where ages were calculated from disturbances occurring before flux observations began. The method used here reflect the current challenges and uncertainties in investigating forest age with FLUXNET sites, but despite these issues the resulting list of flux sites and quantified ages provides a suitable training dataset for ML models to be used in future studies on forest age.

Field-based surveys provide a route to expand existing FLUXNET site data by accurately characterising tree demography but are labour-intensive and therefore spatially restricted (Battison *et al.* 2024). Satellite-based RS approaches overcome this but lack the spatiotemporal resolution for reliable integration with FLUXNET data due to significant scale mismatches between RS pixels and flux tower footprints. The Global Canopy Atlas is a novel data source which utilises airborne laser scanning (ALS) acquisitions to create an analysis-ready map of canopy height and elevation covering 56,554km^2 at 1m^2 resolution across all major biomes, forming an accurate 3-dimensional map of canopy structures (Fischer *et al.* 2025). To showcase this atlas, Fischer *et al.* (2025) developed a framework to standardise forest turnover quantification, highlighting its potential application in automating the quantification of forest ages altered by disturbances from LiDAR imaging at a spatiotemporal resolution great enough to permit use with FLUXNET data.

The potential of standardised, high resolution, and large scale forest age surveys on the horizon opens up the possibility for ML ET models to include forest age as a spatially continuous input feature, which may enable models trained on FLUXNET data to generalise more accurately across forest age gradients at broad scales.

Machine Learning Model Generalisation

ET models can be calibrated or trained with local or regional data. However, local models may demonstrate strong predictive performance locally but weak performance at other locations, potentially rendering application outside of the training area unusable. Therefore, for ML models trained on specific flux tower data, it is important to assess the generalisation power against towers excluded from the training data. ML models trained with data from multiple towers or across a regional scale can be useful for predictions at data-limited sites within a particular region or ecosystem, and have demonstrated more stable performance across multiple sites, albeit with slightly lower accuracy in ecosystems well represented by locally-trained models (Ferreira *et al.* 2021).

The limited and uneven spatial distribution of flux tower sites in the FLUXNET highlights the necessity for robust models which are able to extend ET estimations across a diverse range of landscapes and climatic conditions. Models with better generalisation capabilities should improve estimations for ET in regions with sparse EC - many of which are disproportionately affected by forest disturbances such as fire and pest outbreak from climate change (Altman *et al.* 2024; Haiyang Shi *et al.* 2024). Given the potential impact of these ET estimations on land systems governance, understanding how the inclusion of new variables such as forest age affects model generalisation is crucial before these models can be confidently applied at broader scales.

Conclusion

These findings suggest that the age of a forest represents an underused but potentially important variable in ET estimation models, particularly in complex closed-canopy systems. Age-driven processes such as canopy architecture, rainfall interception, and changes to hydraulic and stomatal conductance introduce complexity to ET dynamics which may not be fully captured by variables used in current models. As forest age distributions become increasingly heterogeneous, the omission of forest age as an input variable for ET models may be limiting their capacity to generalise across successional forest stages.

This project will attempt to quantify the effects of including forest age as a predictor on the generalisation of ET models trained on FLUXNET forest sites, using ML models for their ability to capture both linear and nonlinear effects across large datasets. Crucially, improved predictions are only meaningful when grounded in the biophysical processes underpinning ET and used to improve our understanding of how these may shift under increasing disturbances such as climate change. Without this context, model prediction accuracy is constrained to a technical metric with limited ecological relevance.

Project Aims and Objectives

The overall aims of this project are to investigate the effects of including forest age as a variable in ET ML models and quantify any effects on the generalisation capability of these models across forests of different ages. To achieve this, ML models to estimate ET will be formed using FLUXNET data with different combinations of training and required input variables. After initial data acquisition, the project can be broadly split into four parts:

Part 1

Identify forest FLUXNET sites which contain measurements of suitable variables and are able to have forest age determined, or have already had forest ages quantified. Furthermore, source LAI measurements using flux tower site coordinates. This is crucial as only these sites will be available for training and testing models with forest age and LAI included.

Part 2

Using available FLUXNET data and forest age estimates, appropriate ML architectures will be selected for use with forest age data based on predictive performance against ground truths before the main comparisons. This will help determine which ML methods are suitable for use with forest age data to form accurate predictions to be taken forward.

Part 3

Models will be trained on forests of a certain age and tested against ground truths of a similar forest which is a significantly different age (ideally situated in the same region and containing the same species of trees). Evaluating the accuracy of a model trained on one age and tested on another will aim to quantify the effect forest age has on ET predictions across successional stages, and will inform the design and interpretation of part 4.

Part 4

A four model comparison will be carried out to identify the individual and combined effects of forest age and LAI on ET accuracy. The four models, M_1, M_2, M_3, and M_4, will be designed using identical hyperparameter setups, architectures, and trained on the same sites, differing only in their training and input features as follows:

$$ET_t = M_1\left(X_t\right) \tag{1}$$

$$ET_t = M_2\left(X_t, Age_{forest}\right) \tag{2}$$

$$ET_t = M_3\left(X_t, LAI_t\right) \tag{3}$$

$$ET_t = M_4\left(X_t, Age_{forest}, LAI_t\right) \tag{4}$$

where ET_t is the estimated ET over period t (for example, hourly or daily) and X_t represents the required set of input features over t which all models will use for estimations. LAI is included here as a covariate to test if forest age improves predictive performance beyond LAI for ET ML models. All models will then predict ET against identical held out sites with comparisons of accuracy used to isolate any contributions of forest age and LAI to ET estimations across forests of different successional ages.

The main objective is to determine if forest age is a variable which should be controlled for or included within future ML ET models. The results from this project could help guide researchers in this area to improve the generalisation capabilities of future ML ET models across forest ecosystems.

Project Plan

Data Acquisition

FLUXNET2015 data will initially be downloaded for the age-quantified sites included in the supplementary materials for Simon Besnard *et al.* (2018), as this paper will act as the primary age source for the project. These FLUXNET data sites along with the supplementary materials will contain the variables contained within X_t and Age_{forest} in eqs. (1) to (4). LAI data will be extracted from Terra MODIS instrument data (Myneni *et al.* 2015) using the coordinates of the flux tower sites contained within the FLUXNET data.

Site Selection

For this project, only forest sites will be considered, and so the data will be filtered to only contain these ecosystems (any forest type).

Certain sites implicitly mention age through burn year in FLUXNET metadata, but these will require further verification to quantify their age before being included in this project, for example finding publications on burn dates, as in Simon Besnard *et al.* (2018).

Analysis

Feature Engineering

ML Model Setup and Training

ML Model Evaluation

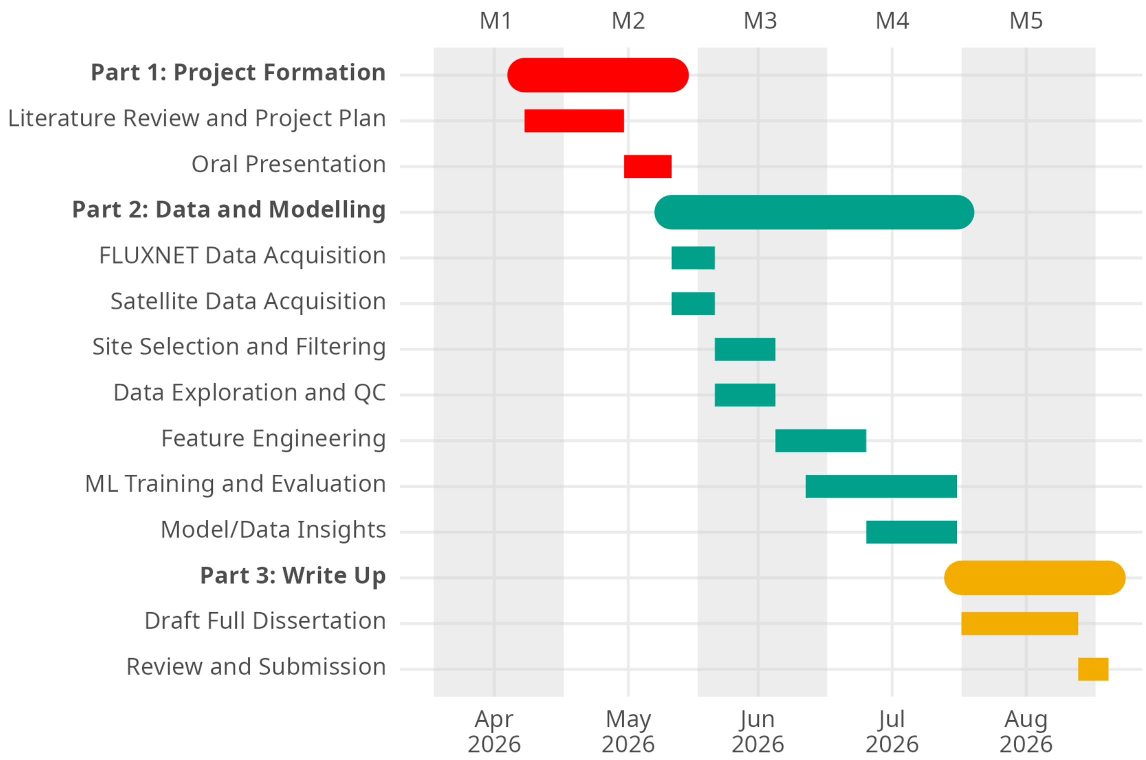
ML Model Predictions

Tools

Interpretation

Gantt Chart

A Gantt chart depicting the expected timeline of key project aspects.



Risk Register ($\sim$ 250 words)

Table 1: Risk register

Nature of the risk	Likelihood and potential impact on the project	Mitigation strategy

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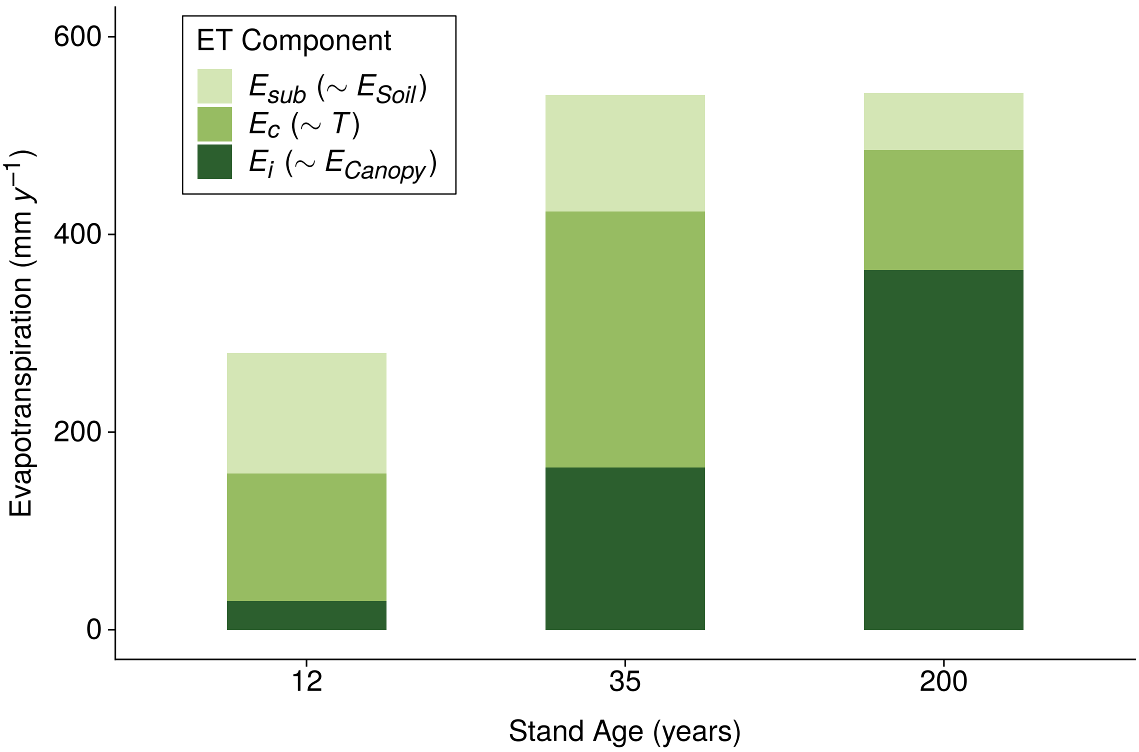

Figure 1: Components of annual evapotranspiration by stand age, adapted from Oishi *et al.* (2020). Sub canopy evapotranspiration (E_{sub}), canopy transpiration (E_c), and canopy interception (E_i) correspond approximately to soil evaporation (E_{Soil}), transpiration (T), and canopy evaporation (E_{Canopy}), with E_{sub} additionally capturing under canopy transpiration.

Table 2: Feature selection of ML-based ET models in the literature

Primary Model(s)	Training Target	Model Input Features	Ecosystem
Random Forest, ANN, cubist, SVM, DBN (T. Xu *et al.* 2018).	Daily ET from 36 EC flux tower sites.	R_s, LAI, RH, T_a (daily averages). P (cumulative prior 30 days).	Mixed: grassland, cropland, ENF, BF, shrubland, wetland, barren land (Heihe River Basin, NW China).
XGboost, ANN (Y. Liu *et al.* 2021).	Daily ET from 17 flux tower sites.	T_a, P, C_a, SW, VPD, NDVI (daily averages). EVI, NIRv, SWIR (time between measurements not defined).	Cropland only.
HS-LSTM (Yan *et al.* 2023).	Daily ET_0 calculated from FAO-56 PM as ground truth.	$T_{a,max}$ and $T_{a,min}$ (for ET_0 from HS which is used as LTSM model input.	Not specified: Semi-humid zone with temperate continental climate (Songliao Basin, NE China).
CatBoost (Ahmadi *et al.* 2024)	Daily ET_0 calculated from FAO-56 PM as ground truth.	R_s (hourly)	Mixed: 17 climatic zones including coastal, central valley, and desert (California, USA).

Where: R_s = solar radiation, LAI = leaf area index, P = precipitation, RH = relative humidity, T_a = air temperature, ENF = evergreen needle leaf forest, BF = broadleaf forest, $T_{a,max/a,min}$ = daily maximum/minimum air temperatures, C_a = atmospheric CO_2 concentration, SW = shortwave solar radiation, NDVI = normalised difference vegetation index, EVI = enhanced vegetation index, NIRv = near infrared reflectance of vegetation, SWIR = shortwave infrared band.